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9. The reversible reaction shown below was one of the first to be studied in detail in the gas and liquid phases and in solution.

$$N_2O_4 \rightleftharpoons 2NO_2$$

(a) One study of the liquid mixture showed that it contained 0.714% NO_2 by mass. Calculate how many moles of NO_2 would be present in 25.0 g of this liquid. Give your answer to an appropriate number of significant figures. [2]

$n(NO_2)$ = ... mol

(b) Both N_2O_4 and NO_2 are soluble in a range of solvents and the equilibrium constant, K_c, can be measured in these.

(i) Give the expression for the equilibrium constant, K_c, for this equilibrium, giving its unit if any. [2]

K_c =

Unit = ...

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(ii) The equilibrium constants for the N_2O_4/NO_2 equilibrium measured at room temperature in some different solvents are listed below.

Solvent	Equilibrium constant, K_c (Unit not shown)
CS_2	17.8
CCl_4	8.05
$CHCl_3$	5.53
C_2H_5Br	4.79
C_6H_6	2.03
$C_6H_5CH_3$	1.69

I. Samples of 0.4000 mol of N_2O_4 are dissolved separately in 1 dm^3 of each of these solvents at room temperature and the reactions allowed to reach equilibrium. In one solution the concentration of N_2O_4 present at equilibrium is 5.81×10^{-2} mol dm^{-3}. Find the value of the equilibrium constant in this solution and hence identify the solvent. [3]

Solvent ....

II. The Gibbs free energy change, ΔG, of this reaction is different in different solvents. Explain how the data shows this and state, with a reason, which solvent would have the most negative ΔG value for this reaction. [2]

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(c) Nitrogen dioxide is used in the production of nitric acid. It is produced in two stages.

Stage 1 $4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(g)$ $\Delta H^\theta = -900\ \text{kJ mol}^{-1}$

Stage 2 $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$ $\Delta H^\theta = -115\ \text{kJ mol}^{-1}$

(i) Explain why the use of a catalyst is essential in industrial processes involving exothermic equilibria such as stage 1. [2]

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(ii) A student attempted to perform stage 2 using a temperature of 450 K and a pressure of 5 atm. He obtained only a small yield. Suggest how the yield could be improved, explaining your reasons. [4]

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